Neural Memory and Interval Calibration

Chengxuan YUAN Student ID: 50025581
DSAA 2012, HKUST(GZ)

1 Motivation and research questions

Financial intervals should expand when volatility rises and contract after a temporary shock. A neural forecaster estimates today's scale from past returns; an online calibrator adjusts it using past errors divided by their originally issued scales. These two histories may reinforce or offset each other. Neural control, joint score–threshold adaptation, and retrospective updates already address adaptation [1–3]. I ask: **how do current neural scale response and historical score memory jointly affect calibration at shock entry and recovery?** The study examines (i) context-by-window interactions across phases and shock durations, (ii) their current/archive components under both decomposition orders, and (iii) persistence across training datasets and random seeds.

2 Approach, data, and evaluation

GRU and causal dilated TCN models learn scales from raw past returns via Gaussian negative log-likelihood and validation early stopping. A zero center isolates uncertainty adaptation. For current context a, archive context b, and score window W, the half-width is

$$h_t^{a,b,W} = s_t^a \, Q_{1-\alpha}\Big(\big\{|y_i|/s_i^b\big\}_{i=t-W}^{t-1}\Big), \qquad \alpha = 0.1.$$

Cross $a, b \in \{16, 64\}$ and $W \in \{32, 128\}$: eight configurations sharing weights within each fit. The diagonal $a = b$ is ordinary calibration; crossed contexts diagnose pathways. All predictions use only past observations.

Paired shock/no-shock simulations share baseline innovations and vary duration, magnitude, tails, and stochastic volatility. I measure the context-by-window interaction after subtracting the matched no-shock contrast, retaining both current/archive decomposition orders. Controls include EWMA, fitted GARCH, and rolling/adaptive/strongly adaptive calibration [4]. Entry and recovery are assessed separately using interval score, coverage, width, and simulated oracle diagnostics. Independent training datasets and paired seeds distinguish fixed-model path variation from retraining sensitivity.

Only generated data and public ECB daily log returns are used. The financial pilot trains through 2018, validates in 2019–2020, and treats the already-inspected period from 2021 as exploratory. Hyperparameters use validation data; new methods require fresh, frozen evaluation.

3 Expected contribution, feasibility, and plan

The intended contribution is a phase-specific mechanism explanation separating current neural scale response from archived score denominators, with reproducible effects and explicit failure boundaries. Exploration and a three-dataset replication establish local computational feasibility with mixed results. Following teaching-team feedback, October will consolidate the question, diagnostics, and literature positioning; November will cover justified robustness and public financial analysis. I will finalize the report, slides, and video before December 11. Supporting design and preliminary observations follow in the appendix.

Preprint.

Appendix A Supporting experimental design

An observed return of magnitude two leaves a score of two if its originally issued scale was one, but a score of one if that scale was two. Thus equal current scales need not give equal calibrated intervals. Input length is not effective learned memory. For each frozen fit, historical denominators remain tied to their own causal predictions; the quantile uses rank $k = \lceil (W+1)(1-\alpha) \rceil$. Window changes also alter finite ranks, so they are not a pure history-length intervention.

For a phase, let $E(a, b, W)$ denote event-minus-control mean interval score and define $D(a, b) = E(a, b, 128) - E(a, b, 32)$. The diagonal interaction $J = D(64, 64) - D(16, 16)$ equals either

$$\begin{aligned} &[D(64,64) - D(16,64)] + [D(16,64) - D(16,16)], \\ \text{or } &[D(64,16) - D(16,16)] + [D(64,64) - D(64,16)]. \end{aligned}$$

These current/archive components are computational differences, not unique causal mediation percentages. Their sign is not an absolute model ranking. Interval score is $2h + (2/\alpha)(|y| - h)_+$; coverage and width accompany it. Simulated true scales and event positions enter construction/evaluation only. The ECB pilot uses USD/JPY/GBP/CHF reference rates per EUR; date-block comparisons retain cross-currency dependence. Restricted data are excluded.

Appendix B Preliminary feasibility and remaining analysis

The completed replication uses three independent training/validation datasets, five paired initialization/sampling seeds per dataset, two architectures, and 128 paired test paths per noise family. Seed indices are shared across datasets; this crossed structure is preserved. Fixed-model paired-path intervals are separate from retraining sensitivity. Model–path combinations and shared duration prefixes are not counted as independent replications.

In the Gaussian reference case, the mean GRU-minus-TCN entry interaction contrast retains its sign in all three new datasets, but its crossed sensitivity interval includes zero. Recovery does not consistently replicate. Validation gradients, older-history permutation, and pulse probes provide diagnostics whose predictive usefulness remains unestablished. Training variability and neural volatility stability have prior literature [5, 6]; the intended increment is this specific pathway study, not a new calibration algorithm or coverage theorem. Deliverables include code, paired results, diagnostic figures, and successful and failed replication conditions.

References

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